

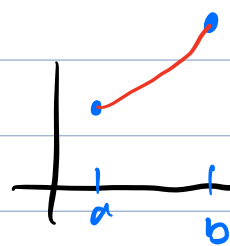
Calc 1 Lecture 18. How derivatives affect the shape of a graph.

What does f' say about f ?

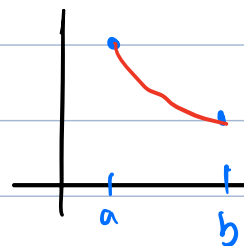
- If $f' > 0$ over an interval, then f is increasing over the interval
- If $f' < 0$ over an interval, then f is decreasing over the interval

Terminology

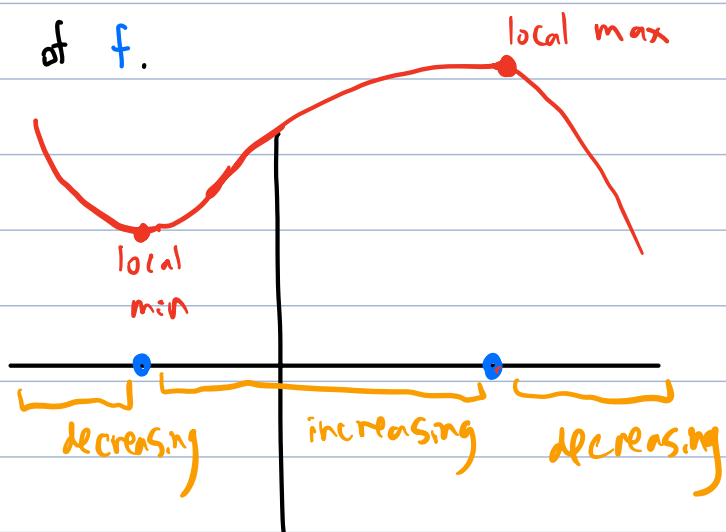
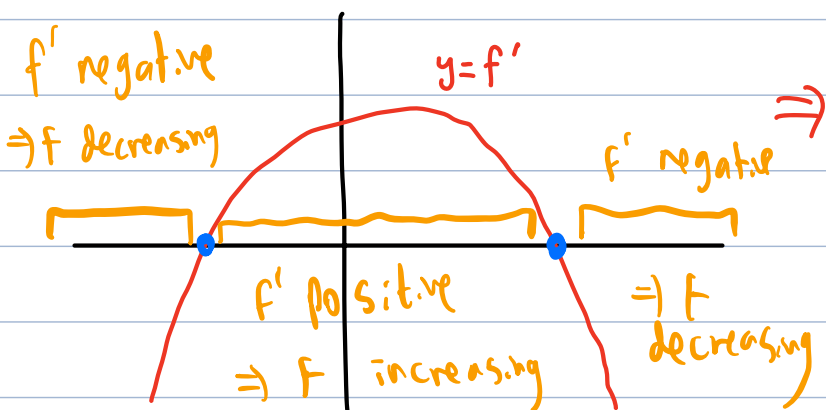
- A function $f(x)$ is increasing over an interval if whenever $a < b$ (two numbers in the interval) then $f(a) < f(b)$



- A function $f(x)$ is decreasing over an interval if whenever $a < b$ (two numbers in the interval) then $f(a) > f(b)$



Example: Consider the following graph of f' . Use it to reconstruct a rough qualitative sketch of the graph of f .



Example Find where the function $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$ is increasing and where it is decreasing.

find $f'(x) = 12x^3 - 12x^2 - 24x$

first find when $f'(x) = 0$: $f'(x) = 12x(x^2 - x - 2)$
 $(x-2)(x+1)$

$f'(x) = 12x(x-2)(x+1) = 0 \Rightarrow x = 0, 2 \text{ or } -1$

when is $f'(x) > 0$?

critical points

when is $f'(x) < 0$?

perform sign analysis...

Approach 1: analyze each factor

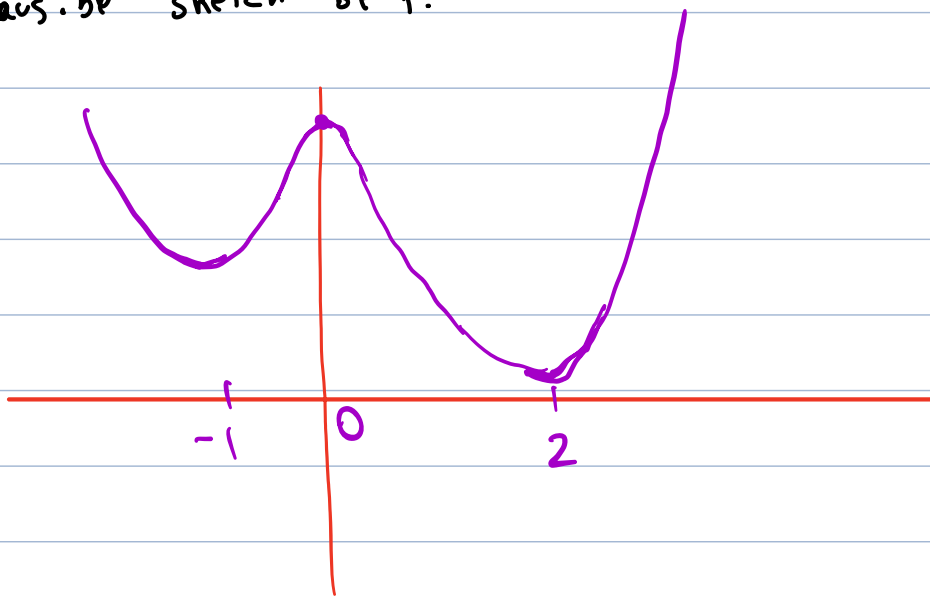
	$x < -1$	-1	$-1 < x < 0$	0	$0 < x < 2$	2	$x > 2$	
x	-		-		+		+	$x > 0$ if $x > 0$
$x-2$	-		-		-		+	$x-2 > 0$ if $x > 2$
$x+1$	-		+		+		+	$x+1 > 0$ if $x > -1$
$f'(x)$	-		+		-		+	

Answer: $f'(x) < 0$ over $(-\infty, -1)$ or $(0, 2)$ } decreasing
 $f'(x) > 0$ over $(-1, 0)$ or $(2, \infty)$ } increasing

Approach 2: Evaluate f' at representative points on each interval.

	-2	-1	-1/2	0	1	2	3
$f'(x)$							

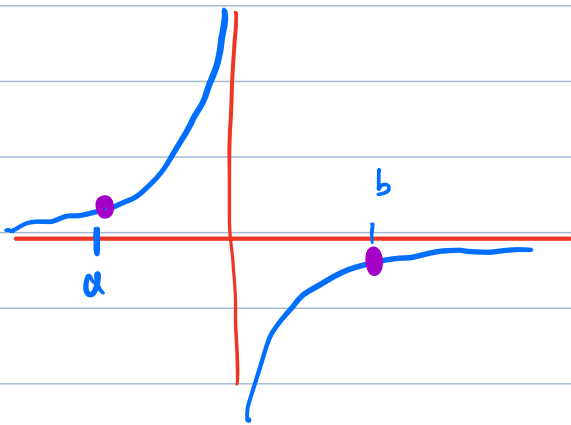
plaus:bb sketch of f:



Cautionary example: Consider $f(x) = -\frac{1}{x}$.

show $f'(x) > 0$ on the entire domain but $a < b$ does not imply $f(a) < f(b)$

$$f'(x) = \frac{d}{dx}(-x^{-1}) = -(-x^{-2}) = x^{-2} = \frac{1}{x^2} > 0$$



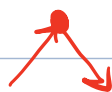
Observation: by analyzing where a function $f(x)$ is increasing and decreasing, we can also deduce which critical points are local min or local max.

The first derivative test Suppose that c is a critical number of a continuous function f . Then:

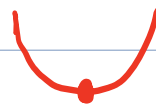
a) If f' changes from positive to negative at c , then f has a local max

to the left of c

to the right of c



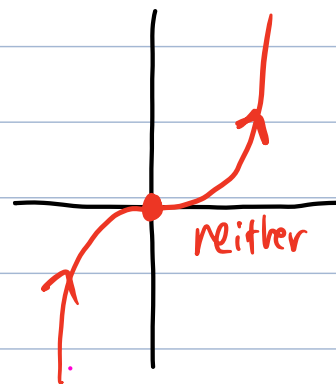
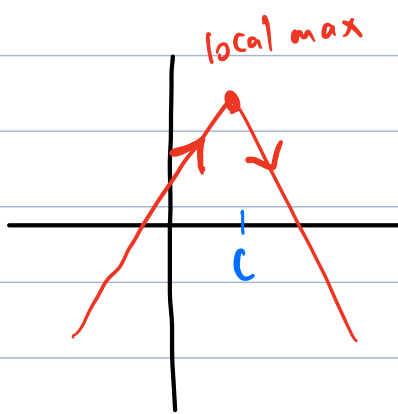
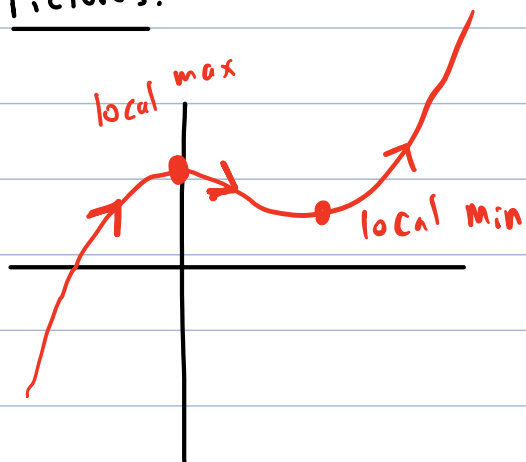
b) If f' changes from negative to positive at c , then f has a local min



c) If f' does not change sign at c , then f has neither a local min nor a local max at c



Pictures:



Example Find the local minimum and maximum values of the function f in the previous example.

value = y value

$$f(x) = 3x^4 - 4x^3 - 12x^2 + 5$$

	$x < -1$	-1	$-1 < x < 0$	0	$0 < x < 2$	2	$x > 2$
x	-		-		+		+
$x-2$	-		-		-		+
$x+1$	-		+		+		+
$f'(x)$	-		+		-		+

Diagram showing the sign of $f'(x)$ and the corresponding local extrema:

- At $x = -1$, $f'(x)$ changes from - to +, indicating a **local min**.
- At $x = 0$, $f'(x)$ changes from + to -, indicating a **local max**.
- At $x = 2$, $f'(x)$ changes from - to +, indicating a **local min**.

local min values

$$= f(-1), f(2)$$

$$= \boxed{0, -27}$$

local max value

$$= f(0) = \boxed{5}$$

Example Find the local maximum and minimum values of the function

$$g(x) = x + 2\sin(x)$$

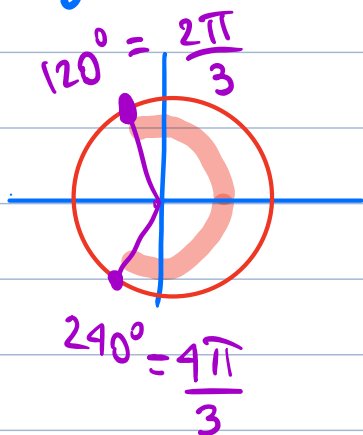
$$0 \leq x \leq 2\pi$$

Step 1: find $g'(x) = 1 + 2\cos(x)$

Step 2: Find critical points

$$g'(x) = 0 \Rightarrow 1 + 2\cos x = 0 \Rightarrow \cos x = -\frac{1}{2}$$

critical points: $\boxed{\frac{2\pi}{3}, \frac{4\pi}{3}}$



Step 3: Sign analysis of g'

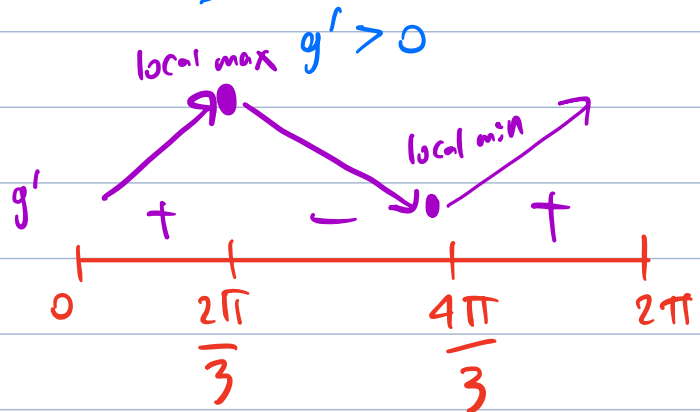
$$g'(x) > 0 \Leftrightarrow 1 + 2\cos(x) > 0 \Rightarrow 2\cos(x) > -1$$

$$\cos(x) > -\frac{1}{2}$$

$$\Rightarrow \boxed{0 < x < \frac{2\pi}{3} \text{ OR } \frac{4\pi}{3} < x < 2\pi}$$

$$g'(x) < 0 \text{ if } \cos x < -\frac{1}{2}$$

$$\Rightarrow \boxed{\frac{2\pi}{3} < x < \frac{4\pi}{3}}$$



$$g(x) = x + 2\sin(x)$$

$$\text{local max value: } g\left(\frac{2\pi}{3}\right) = \frac{2\pi}{3} + 2\sin\left(\frac{2\pi}{3}\right) = \frac{2\pi}{3} + 2 \cdot \frac{\sqrt{3}}{2} = \boxed{\frac{2\pi}{3} + \sqrt{3}}$$

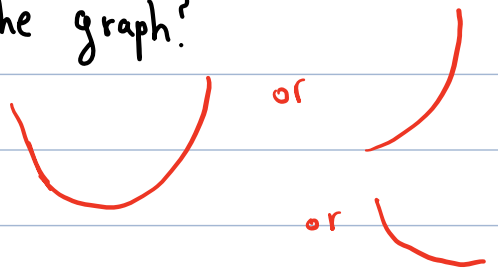
$$\text{local min value: } g\left(\frac{4\pi}{3}\right) = \frac{4\pi}{3} + 2\sin\left(\frac{4\pi}{3}\right) = \frac{4\pi}{3} + 2 \cdot \left(-\frac{\sqrt{3}}{2}\right) = \boxed{\frac{4\pi}{3} - \sqrt{3}}$$

What does f'' say about f ?

• Suppose $f'' > 0$. How is this reflected in the graph?

$$(f')' > 0 \Rightarrow f' \text{ is increasing}$$

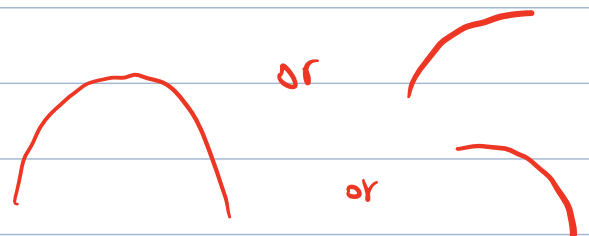
concave up



• Suppose $f'' < 0$. How is this reflected in the graph?

$$(f')' < 0 \Rightarrow f' \text{ is decreasing}$$

concave down



Definition. If the graph of f lies above all of its tangents on an interval, then it is called **concave up** over that interval

• If the graph of f lies below all of its tangents on an interval, then it is called **concave down** over that interval

Concavity test

a) If $f''(x) > 0$ for all x in an interval, then f is concave up over that interval

b) If $f''(x) < 0$ for all x in an interval, then f is concave down over that interval.

if $f''(x) = 0$, inconclusive.

Simple example: Analyze the concavity of $f(x) = x^3 - x$
(in addition to increasing/decreasing)

increasing/decreasing:

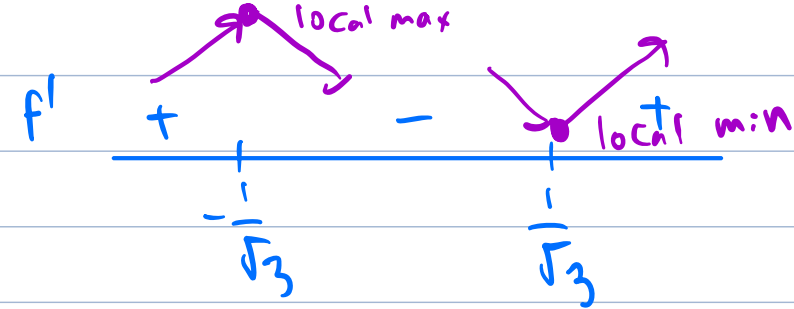
$$f'(x) = 3x^2 - 1$$

critical points: $3x^2 - 1 = 0 \Rightarrow x^2 = \frac{1}{3} \Rightarrow x = \pm \frac{1}{\sqrt{3}}$

$$f'(x) \stackrel{?}{>} 0 \Leftrightarrow 3x^2 - 1 > 0 \Leftrightarrow 3x^2 > 1 \Leftrightarrow x^2 > \frac{1}{3}$$

$$\Leftrightarrow \boxed{x > \frac{1}{\sqrt{3}} \text{ or } x < -\frac{1}{\sqrt{3}}}$$

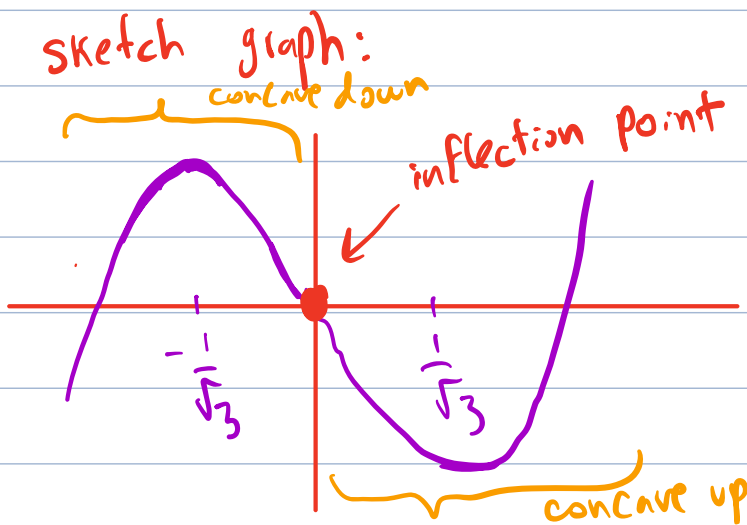
$$f'(x) < 0 \Leftrightarrow 3x^2 < 1 \Leftrightarrow x^2 < \frac{1}{3} \Leftrightarrow \boxed{-\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}}}$$



Concavity:

$$f''(x) = \frac{d}{dx}(3x^2 - 1) = 6x > 0 \text{ if } \boxed{x > 0} \text{ concave up}$$

$$< 0 \text{ if } \boxed{x < 0} \text{ concave down}$$



Definition: A point on a curve $y = f(x)$ is called an inflection point if f is continuous at that point and as you cross that point, the function changes concavity.

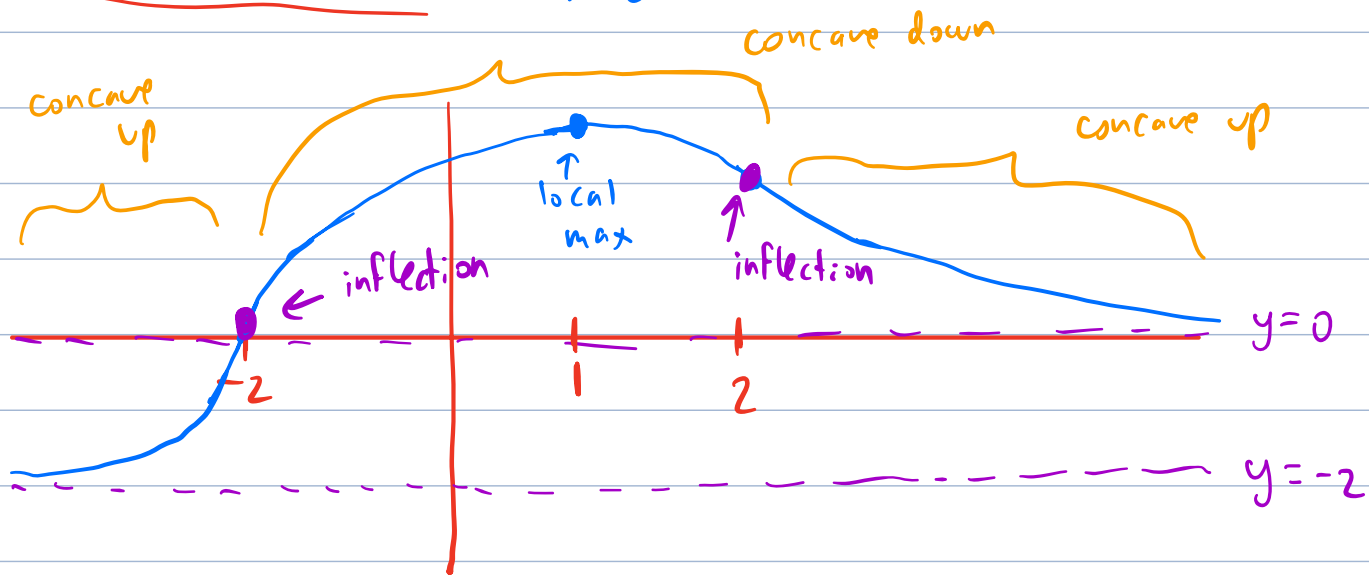
(e.g. $x=0$ is an inflection point in the previous example)

Example sketch a possible graph of a function f that satisfies the following conditions:

i) $f'(x) > 0$ on $(-\infty, 1)$, $f'(x) < 0$ on $(1, \infty)$.

ii) $f''(x) > 0$ on $(-\infty, -2)$ and $(2, \infty)$, $f''(x) < 0$ on $(-2, 2)$

iii) $\lim_{x \rightarrow -\infty} f(x) = -2$, $\lim_{x \rightarrow \infty} f(x) = 0$



The second derivative gives us another way of testing local max/min.

The second derivative test: Suppose f'' is continuous near c , and $f'(c) = 0$.

a) If $f''(c) > 0$, then f has a local min at c

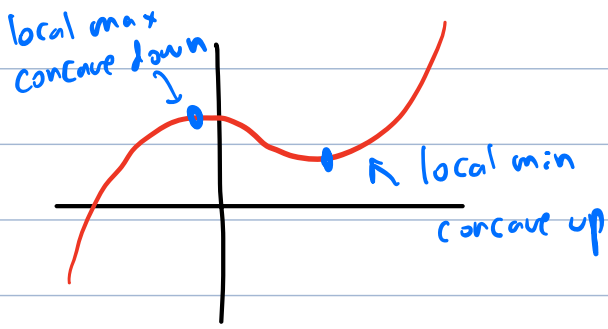


b) If $f''(c) < 0$, then f has a local max at c



Note: if $f''(c) = 0$, inconclusive (don't have enough information)

Picture



Note: we now have two tests for local max/min. What are pros and cons?

1st deriv. test. Pros: • if you're already interested in finding where f is increasing/decreasing then you have the necessary info ready
• more conclusive

2nd deriv. test. pros: • requires less calculation if you're only interested in finding the local min/max

Example Discuss the curve $y = \overbrace{x^4 - 4x^3}^{f(x)}$ with respect to concavity, points of inflection, and local max and min. Use this info to sketch the curve.

$$f'(x) = \underline{4x^3 - 12x^2}$$

$$\text{critical points: } 4x^3 - 12x^2 = 0 \Rightarrow 4x^2(x - 3) = 0$$

$$\Rightarrow \underline{x=0} \text{ or } \underline{x=3}$$

critical points

$$f''(x) = \frac{d}{dx} (4x^3 - 12x^2) = \underline{12x^2 - 24x}$$

$$f''(x) = 0: \quad 12x(x-2) = 0 \Rightarrow \underline{x=0} \text{ or } \underline{x=2}$$

inflection points?

sign analysis of f'' :

	-1	0	1	2	3
$f''(x)$	36		-12		$12 \cdot 3 \cdot 1 = 36$
	+		-		+

concave up on $(-\infty, 0) \cup (2, \infty)$

concave down on $(0, 2)$

inflection points: $x=0, x=2$

local min/max?

$x=0$ or $x=3$
critical points

try 2nd deriv. test.

$$f''(3) = 36 > 0 \Rightarrow \underline{\text{local min}}$$

$$f''(x) = 12x^2 - 24x$$

$$f''(0) = 0 \Rightarrow \text{inconclusive}$$

$\Rightarrow$ use first deriv. test instead.

$$f'(x) = 4x^3 - 12x^2 = 4x^2(x-3)$$

sign analysis of f' near $x=0$:

if $x < 0$: $f'(x) = \underbrace{4x^2}_{\text{pos}} \underbrace{(x-3)}_{\text{neg}} = \underline{\text{neg}}$

pos neg

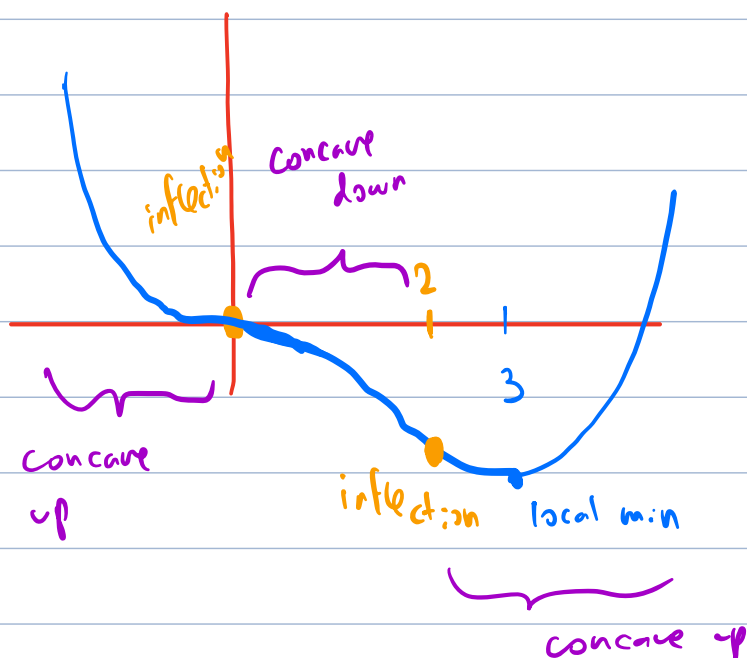
$$\text{if } 0 < x < 3: f'(x) = \underbrace{4x^2}_{\text{pos}} \underbrace{(x-3)}_{\text{neg}} = \underline{\text{neg}}$$

So $x=0$ is neither a local min nor a local max. $\searrow \swarrow$

Sketch of curve:

critical points $\underline{0}, \underline{3}$
neither local min

inflection points $0, 2$



More examples

Example sketch the graph of the function $f(x) = x^{2/3} (6-x)^{1/3}$ by analyzing $f'(x)$

$$\begin{aligned} f'(x) &= \frac{d}{dx} (x^{2/3}) (6-x)^{1/3} + x^{2/3} \frac{d}{dx} ((6-x)^{1/3}) \\ &= \frac{2}{3} x^{-1/3} (6-x)^{1/3} + x^{2/3} \cdot \frac{1}{3} (6-x)^{-2/3} \cdot \overbrace{\frac{d}{dx} (6-x)}^{-1} \\ &= \frac{2}{3} x^{-1/3} (6-x)^{1/3} - x^{2/3} \cdot \frac{1}{3} (6-x)^{-2/3} \end{aligned}$$

simplify...

$$\begin{aligned} &= \frac{2(6-x)^{1/3}}{3 x^{1/3}} - \frac{x^{2/3}}{3 (6-x)^{2/3}} \\ \text{common} & \quad \overbrace{(6-x)^{1/3} = 6-x} \quad \overbrace{x^{1/3} = x} \\ \text{denom.} & \quad = \frac{2(6-x)^{1/3} \cdot (6-x)^{2/3}}{3 x^{1/3} (6-x)^{2/3}} - \frac{x^{2/3} \cdot x^{1/3}}{3 (6-x)^{2/3} x^{1/3}} \end{aligned}$$

$$= \frac{2(6-x)}{3 x^{1/3} (6-x)^{2/3}} - \frac{x}{3 x^{1/3} (6-x)^{2/3}} \quad f'(x)$$

$$= \frac{2(6-x) - x}{3 x^{1/3} (6-x)^{2/3}} = \frac{12 - 3x}{3 x^{1/3} (6-x)^{2/3}} = \boxed{\frac{4-x}{x^{1/3} (6-x)^{2/3}}}$$

critical points:

$$f'(x) = 0 : \text{numerator} = 0 \Rightarrow 4-x=0 \Rightarrow \boxed{x=4}$$

$$f'(x) \text{ undefined} : \text{denominator} = 0 \Rightarrow \boxed{x=0, x=6}$$

sign analysis of f'

	0	4	6
$4-x$	+	+	-
$x^{1/3}$	-	+	+
$(6-x)^{2/3}$	+	+	+
f'	-	+	-

$$4-x > 0? \Leftrightarrow x < 4$$

$$x^{1/3} > 0? \Leftrightarrow x > 0$$

$$(6-x)^{2/3} \geq 0 \text{ always}$$

$$\left(\sqrt[3]{6-x} \right)^2 \geq 0$$



local
min

$$x=0$$

cusp



local
max

$$x=4$$

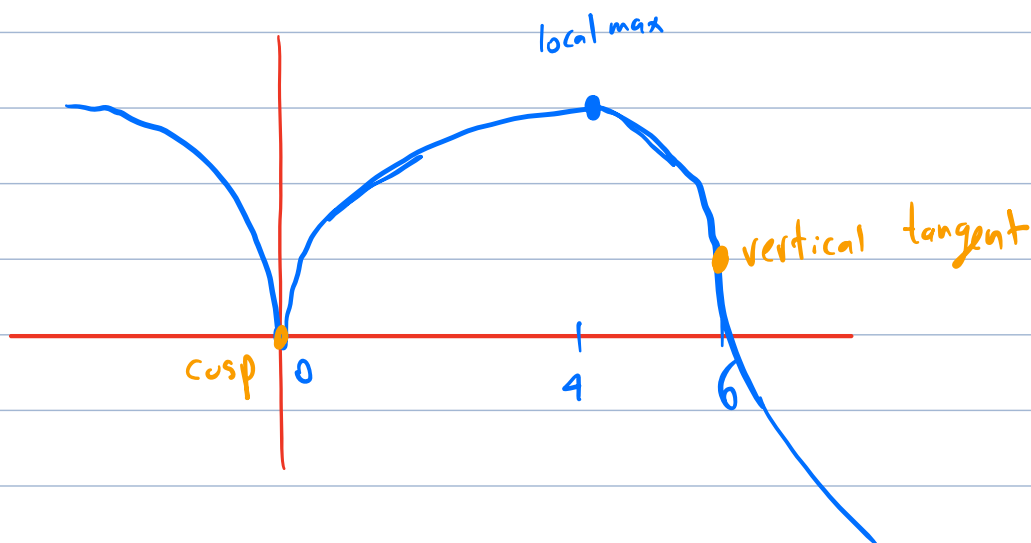


neither

$$x=6$$

vertical tangent

sketch of function



Example sketch the graph of $e^{1/x}$ using its first and second derivatives and asymptotes.
vertical and horizontal

VA: $x=0$

$$\lim_{x \rightarrow 0^+} (e^{1/x}) = e^{\infty} = \infty$$

$$\lim_{x \rightarrow 0^-} (e^{1/x}) = e^{-\infty} = 0$$

HA: $\lim_{x \rightarrow \infty} e^{1/x} = e^0 = \underline{1}$

$$\lim_{x \rightarrow -\infty} e^{1/x} = e^0 = \underline{1}$$

$$f(x) = e^{1/x}$$

$$f'(x) = \underbrace{e^{1/x}}_{\text{pos}} \cdot \underbrace{-\frac{1}{x^2}}_{\text{neg}} = \text{negative}$$

concavity: $f''(x) = \frac{d}{dx} \left(-\frac{e^{1/x}}{x^2} \right) = - \frac{\overbrace{\frac{d}{dx}(e^{1/x})}^{-\frac{e^{1/x}}{x^2}} \cdot \overbrace{x^2}^{2x} - e^{1/x} \cdot \overbrace{\frac{d}{dx}(x^2)}^{2x}}{\underbrace{(x^2)^2}_{x^4}}$

$$= - \frac{\cancel{\frac{-e^{1/x}}{x^2}} \cdot \cancel{x^2} - e^{1/x} (2x)}{x^4} = \frac{e^{1/x} + e^{1/x} \cdot 2x}{x^4}$$

$$= \boxed{\frac{e^{1/x} \cdot (1+2x)}{x^4}} \quad f''$$

$$f''(x) = \frac{e^{1/x} \cdot (1+2x)}{x^4}$$

$\overset{\text{pos}}{\circledast} e^{1/x}$
 $\overset{\text{pos}}{\circledast} x^4$

pos. if $x > -\frac{1}{2}$ concave up

neg. if $x < -\frac{1}{2}$ concave down

inflection at $x = -\frac{1}{2}$

Sketch:

